



Experimental investigation of shaping disturbance observer design for motion control of precision mechatronic stages with resonances

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ABSTRACT

In this paper, shaping disturbance observer (SDOB) is investigated for precision mechatronic stages with middle-frequency zero/pole type resonance to achieve good motion control performance in practical manufacturing situations. Compared with traditional standard disturbance observer (DOB), in SDOB a pole-zero cancellation based shaping filter is cascaded to the mechatronic stage plant to meet the challenge of motion control performance deterioration caused by actual resonance. Noting that pole-zero cancellation is inevitably imperfect and the controller may even consequently become unstable in practice, frequency domain stability analysis is conducted to find out how each parameter of the shaping filter affects the control stability. Moreover, the robust design criterion of the shaping filter, and the design procedure of SDOB, are both proposed to guide the actual design and facilitate practical implementation. The SDOB with the proposed design criterion is applied to a linear motor driven stage and a voice motor driven stage, respectively. Experimental results consistently validate the effectiveness nature of the proposed SDOB scheme in practical mechatronics motion applications. The proposed SDOB design actually could be an effective unit in the controller design for motion stages of mechanical manufacture equipments.

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1. Introduction

Precision mechatronic motion stages are widely used in modern mechanical manufacture equipments, such as machine tools and IC lithography tools [1–3]. Continuously rigid requirements on motion accuracy and speed make the motion control technology become more and more important [4,5]. To achieve the desired motion control performance, disturbance observer (DOB) is widely used as an inner loop controller unit to reject disturbance and improve the motion control performance [6–8]. One main advantage of the DOB is that it is an add-on inner disturbance rejection part with respect to the outer feedback controller and almost has no influence on the nominal performance of the system [9,10]. To some extent, the DOB and outer feedback controller could be designed separately, which leads that advanced DOB design method becomes an important issue to facilitate the controller design.

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In DOB, the inverse of the nominal model as well as a low-pass filter, i.e., Q , is used to estimate the disturbance of the plant system. The cut-off frequency of Q is limited by the un-modelled dynamics of the mechatronic motion plant in order to maintain robust stability for the DOB inner loop [11]. Xu [12] proposed a novel scheme of plant shaping DOB (SDOB) which utilized several notch filters to cancel the plant's major resonances. Herein, the notch filters were designed through optimization so that the shaped plant dynamics could approximate the nominal model.

Inspired by the characteristics of SDOB way, this paper studies the practical performance oriented design method of SDOB instead of mathematical optimization specially for mechatronic motion system with middle frequency zero/pole type resonance. This kind of resonance is common for mechatronic motion stages where the 1st-order modal frequency is usually not high (the reason is explained in Section 3.1). The resonance is cancelled by its inverse dynamics, i.e., pole-zero cancellation. And the stability is discussed when there exist uncertainties in resonance parameters. Moreover, the robust design criterion of SDOB shaping filter is proposed to facilitate actual implementation. The proposed SDOB design procedure is also presented to facilitate implementation. Finally, the SDOB with the proposed design criterion is applied to a linear motor driven stage and a voice motor driven stage, respectively. Comparative experimental results consistently verify the effectiveness nature of the proposed SDOB scheme in practical motion control applications. The proposed SDOB design actually provides an effective unit and could be well integrated with manual loop shaping design of the outer feedback controller for motion stages of manufacture equipments.

The remainder of this paper is organized as follows. Section 2 provides a brief review of standard DOB and SDOB. Section 3 discusses the robust performance of SDOB with respect to uncertainty of resonances through simulation analysis, and presents a robust design rule for SDOB. Section 4 describes the controller design for precision mechatronic motion stages with experimental investigation. The conclusions are drawn in Section 5.

2. DOB and SDOB

2.1. Standard disturbance observer

As computer control systems practically run digital controllers, the theory of discrete DOB is briefly described in the following [11,12]. Fig. 1 shows the block diagram of DOB based control loop, where $C_f(z^{-1})$ is the feedback controller, $G_p(z^{-1})$ is the plant, and the part inside the box with gray background is DOB. DOB includes the low pass filter $Q(z^{-1})$, the inverse model $[G_n^*(z^{-1})]^{-1}$, and the delay z^{-m} . $G_n^*(z^{-1})$ approximates the part in the actual plant apart from time delay (denoted by $G_p^*(z^{-1})$), and

$$G_p(z^{-1}) = z^{-m}G_p^*(z^{-1}) \quad (1)$$

The DOB loop in Fig. 1 inside the gray box could be transformed into its equivalent form as showed in Fig. 2. Then the transfer functions from the command, the disturbance and the noise to the plant output are

$$G_{yc}(z^{-1}) = \frac{z^{-m}G_p^*(z^{-1})G_n^*(z^{-1})}{G_n^*(z^{-1}) + Q(z^{-1})[z^{-m}G_p^*(z^{-1}) - z^{-m}G_n^*(z^{-1})]} \quad (2)$$

$$G_{yd}(z^{-1}) = \frac{z^{-m}G_p^*(z^{-1})G_n^*(z^{-1})[1 - z^{-m}Q(z^{-1})]}{G_n^*(z^{-1}) + Q(z^{-1})[z^{-m}G_p^*(z^{-1}) - z^{-m}G_n^*(z^{-1})]} \quad (3)$$

$$G_{yn}(z^{-1}) = -\frac{z^{-m}G_p^*(z^{-1})Q(z^{-1})}{G_n^*(z^{-1}) + Q(z^{-1})[z^{-m}G_p^*(z^{-1}) - z^{-m}G_n^*(z^{-1})]} \quad (4)$$

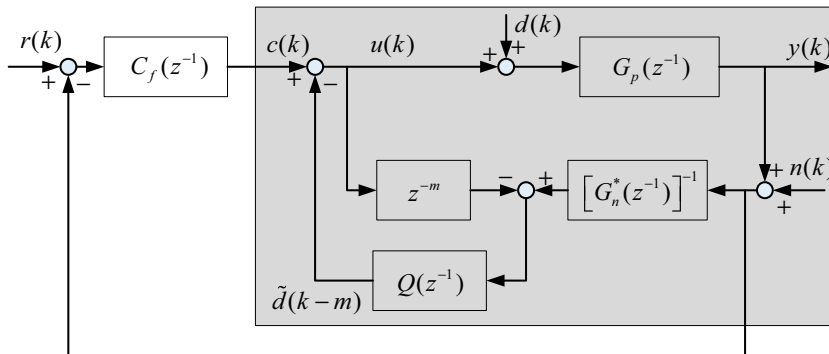


Fig. 1. Block diagram of DOB control loop.

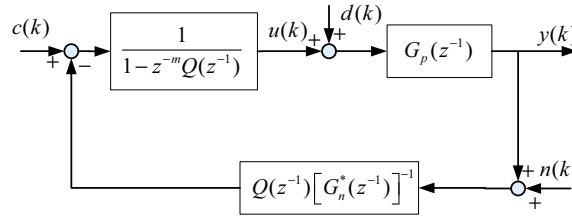


Fig. 2. Equivalent diagram of DOB.

where $Q(z^{-1})$ has low pass property so that $Q(z^{-1}) \rightarrow 1$ at low frequency and $Q(z^{-1}) \rightarrow 0$ at high frequency. Therefore, for the DOB loop transfer function, $G_{yc} \rightarrow G_n$ at low frequency and $G_{yc} \rightarrow G_p$ at high frequency; for the disturbance, $G_{yd} \rightarrow 0$ at low frequency and $G_{yd} \rightarrow G_p$ at high frequency; for the noise, $G_{yn} \rightarrow -1$ at low frequency and $G_{yn} \rightarrow 0$ at high frequency. The above properties cause that DOB has good disturbance rejection capability with limited influence on the outer feedback controller.

Noting the expressions (2) and (3), it can be found that different characteristics for Q are needed by the requirements of keeping the inner loop dynamics unchanged and good disturbance rejection ability. The former demands Q to be a low pass filter. However, for the later, it is better to design $(1 - Q)$ as a high pass filter. Another function of Q is to make $Q[G_n^*]^{-1}$ causal because there usually exists one double integrator in the motion stages dynamics. The following optimal Q could achieve a balance between the above requirements

$$Q(s) = \frac{3\tau s + 1}{\tau^3 s^3 + 3\tau^2 s^2 + \tau s + 1} \quad (5)$$

The bandwidth of Q is determined by τ , and the discrete form $Q(z^{-1})$ could be obtained by zero-order-hold (ZOH) or Tustin transformation.

Let $\Delta(z^{-1})$ be the multiplicative uncertainty that describes the mismatch between the nominal model and the actual plant model, i.e.,

$$G_p(z^{-1}) = z^{-m} G_p^*(z^{-1}) = z^{-m} G_n^*(z^{-1}) [1 + \Delta(z^{-1})] \quad (6)$$

The sufficient condition for robust stability of the DOB loop is given in the following without detailed derivation, i.e.,

$$|\Delta(e^{j\omega T_s}) Q(e^{j\omega T_s})| < 1, \quad \forall \omega \quad (7)$$

The bandwidth of Q should be maximized to have good disturbance rejection ability without violating the robust condition in (7).

2.2. Shaping disturbance observer

In practice, mechatronic motion systems usually have modes due to structural flexibility, which causes resonances in the control system models at high or middle frequency. In DOB design, the resonances would increase the model uncertainty Δ , and limit the bandwidth of Q resultantly if the robustness is maintained. SDOB provides a rather simple but effective method to solve that problem when the plant model has resonances. Fig. 3 shows the diagram of SDOB. Herein, a shaping filter

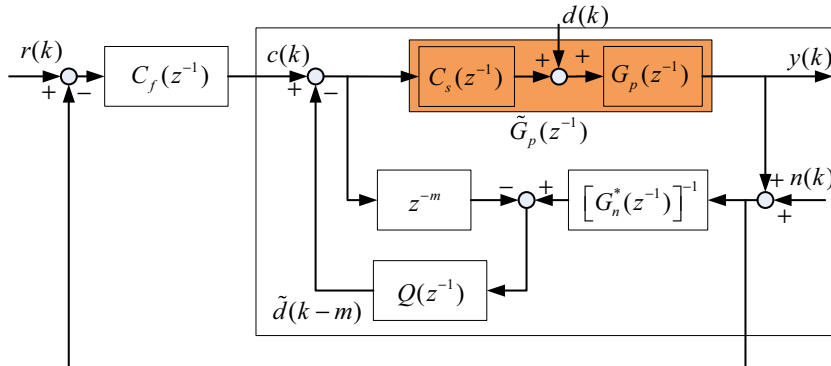


Fig. 3. Block diagram of SDOB control loop.

$C_s(z^{-1})$, different from the notch filter in [12], is used to attenuate the resonances characteristics of the plant, and the shaped plant is

$$\tilde{G}_p(z^{-1}) = G_p(z^{-1})C_s(z^{-1}) \quad (8)$$

The time delay part could be estimated by the phase characteristics of the identified model, and the rest part could be fitted by Matlab function `fitfrd()`.

If the shaping filter $C_s(z^{-1})$ is carefully designed so that the resonances have been cancelled well, $\tilde{G}_p(z^{-1})$ could be treated as a new plant model without resonance. And a simple nominal plant model could still be used to satisfy

$$\tilde{G}_p(z^{-1}) = z^{-m}G_p^*(z^{-1})C_s(z^{-1}) \approx z^{-m}G_n^*(z^{-1}) \quad (9)$$

Then the DOB method could be used for the design of SDOB without difficulties. In SDOB, the input-output dynamics and robust condition are similar to those in (2)–(4) and (7) respectively.

For the design of shaping filter $C_s(z^{-1})$, a mathematical optimization method is proposed in [12]. However, to accommodate the idea of loop shaping, inverse model of the resonances is used as the shaping filter concisely.

The resonances in the mechatronic plant model can be expressed as

$$G_{res}(s) = \prod_{i=1}^n \frac{\left(\frac{s}{\omega_{z,i}}\right)^2 + 2\zeta_{z,i}\frac{s}{\omega_{z,i}} + 1}{\left(\frac{s}{\omega_{p,i}}\right)^2 + 2\zeta_{p,i}\frac{s}{\omega_{p,i}} + 1} \quad (10)$$

where $\zeta_{z,i}$, $\omega_{z,i}$, $\zeta_{p,i}$, $\omega_{p,i}$ are the damping ratio and natural frequency of the zero and pole for the i th mode, respectively. The shaping filter can be chosen as

$$C_s(s) = \prod_{i=1}^n \frac{\left(\frac{s}{\omega_{z,i}^s}\right)^2 + 2\zeta_{z,i}^s\frac{s}{\omega_{z,i}^s} + 1}{\left(\frac{s}{\omega_{p,i}^s}\right)^2 + 2\zeta_{p,i}^s\frac{s}{\omega_{p,i}^s} + 1} \quad (11)$$

The parameters of C_s should be set as $\omega_{z,i}^s = \omega_{p,i}$, $\zeta_{z,i}^s = \zeta_{p,i}$, $\omega_{p,i}^s = \omega_{z,i}$, $\zeta_{p,i}^s = \zeta_{z,i}$, so that the resonances can be perfectly canceled. Herein, the transfer function is expressed in s -domain for the sake of convenience, and the discrete $C_s(z)$ could be designed with the aid of w -transformation in w plane, which is similar to s plane [13] or Matlab SISO Toolbox.

3. Robustness analysis of SDOB

3.1. Robustness analysis of shaping filter parameters

The issue of robustness arises due to the fact that, in practical implementation it is difficult to accurately obtain the parameters of the plant model by identification, especially for the damping ratios. The parameters of the resonances also may drift slightly because of temperature variation, components aging, etc.

As the rigorous robust analysis for shaping filter that subjects to general cases is rather difficult, in this paper emphasis is placed on the important and common cases those make sense in practice. First of all, middle frequency resonance (usually the 1st-order resonance) has much more impacts on stability than high frequency resonances. On the other hand, there are two major types resonances in motion stages, i.e., the zero/pole resonance ($\omega_{z,i} < \omega_{p,i}$) and pole/zero resonance ($\omega_{z,i} > \omega_{p,i}$), which are determined by the structural characteristics of the stages. When the resonance locates in the middle frequency range, the former has phase lead and will not harm the close loop stability for the feedback controller, but the later causes phase lag and will threaten the stability. When the 1st-order resonance of mechatronic motion stages is not high enough, i.e., becomes the middle frequency resonance, it is usually a good choice to make the resonance to be zero/pole type by proper structural design [14]. Therefore, this paper only concerns the case of mechatronic plant with middle frequency zero/pole resonance.

To perform robustness analysis for the parameters of the shaping filter, a virtual plant with middle frequency zero/pole resonance is constructed, and feedback controller and SDOB are designed for the plant. The details of the plant model and the controller are showed in Appendix. The robust performance analysis is not performed on control loop with perturbed plant model and fixed-parameter controller. On the contrary, perturbations are applied on each parameter for the shaping filter on the basis of the ideal shaping filter (denoted as $C_s^0(z)$) while the plant model is fixed. The two methods are equivalent for the evaluation of robustness, but the latter will help to obtain an empirical robust design rule for the design of shaping filter. $C_s^0(z)$ cancels the 1st-order resonance of the plant, and the parameters are $\omega_z^{shp} = \omega_{p,1} = 390$ rad/s, $\zeta_z^{shp} = \zeta_{p,1} = 0.1$, $\omega_p^{shp} = \omega_{z,1} = 314$ rad/s, $\zeta_p^{shp} = \zeta_{z,1} = 0.1$. Figs. 4 and 5 show the Nyquist plots of the overall open loop transfer function ($L = G_{yc}C$) when one of the four parameters changes, while the others equal to their ideal values. In the figures, the changing

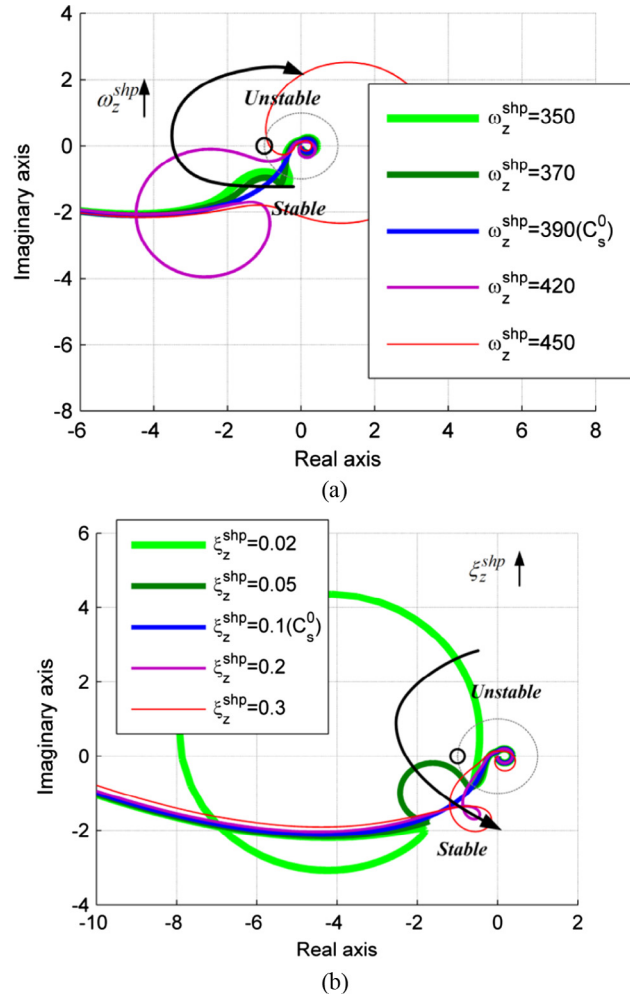


Fig. 4. Nyquist plots and stability situations for shaping filters with different zero parameters. (a) Influences of the zero frequency; (b) Influences of the zero damping ratio.

of Nyquist locus clearly demonstrates how each parameter of $C_s(z)$ influences the stability of the control loop. Fig. 4 shows the influence of parameters of the zero, while Fig. 5 shows the situations for the pole parameters.

Fig. 4(a) shows that when ω_z^{shp} become larger than its ideal value, the Nyquist locus gets closer to the point $(-1, 0j)$ and eventually encircle it, and the Nyquist locus does not approach or encircle the point $(-1, 0j)$ when ω_z^{shp} is smaller. In Fig. 4(b), the Nyquist locus approaches and eventually encircle $(-1, 0j)$ when ζ_z^{shp} becomes larger than the ideal value, while the Nyquist locus moves to the opposite direction and no encirclement happens. However, it seems that when ζ_z^{shp} becomes too large, encirclement of $(-1, 0j)$ may happen at higher frequency segment. Therefore, let the shaping zero frequency ω_z^{shp} be smaller and the shaping zero damping ζ_z^{shp} larger slightly than their ideal values ($\omega_z^{shp} < \omega_{p,1}$ and $\zeta_z^{shp} > \zeta_{p,1}$), and that would gain some robustness with respect to the uncertainties of resonance pole parameters. Similar conclusions could be drawn for shaping pole parameters from Fig. 5. When ω_p^{shp} gets smaller or ζ_p^{shp} gets larger, the stability margins decrease and instability may occur, and the stability margins increase when ω_p^{shp} gets larger or ζ_p^{shp} gets smaller. But it seems that stability is not very sensitive to the shaping pole parameters, especially for ζ_p^{shp} , and control loop may keep stable when ζ_p^{shp} varies within a large range. Considering that small pole damping ratio ζ_p^{shp} will cause large controller output around pole frequency, thus ζ_p^{shp} could be a little larger than its ideal value ($\zeta_p^{shp} > \zeta_{z,1}$). Also the resonance frequencies could be precise and have fewer uncertainties, the frequencies of shaping filter could be set to the ideal values to reduce the loss of design performance. Therefore, the value design rule for the shaping filter could be summarized as $\omega_z^{shp} \leq \omega_{p,1}$, $\zeta_z^{shp} > \zeta_{p,1}$, $\omega_p^{shp} \geq \omega_{z,1}$, $\zeta_p^{shp} > \zeta_{z,1}$.

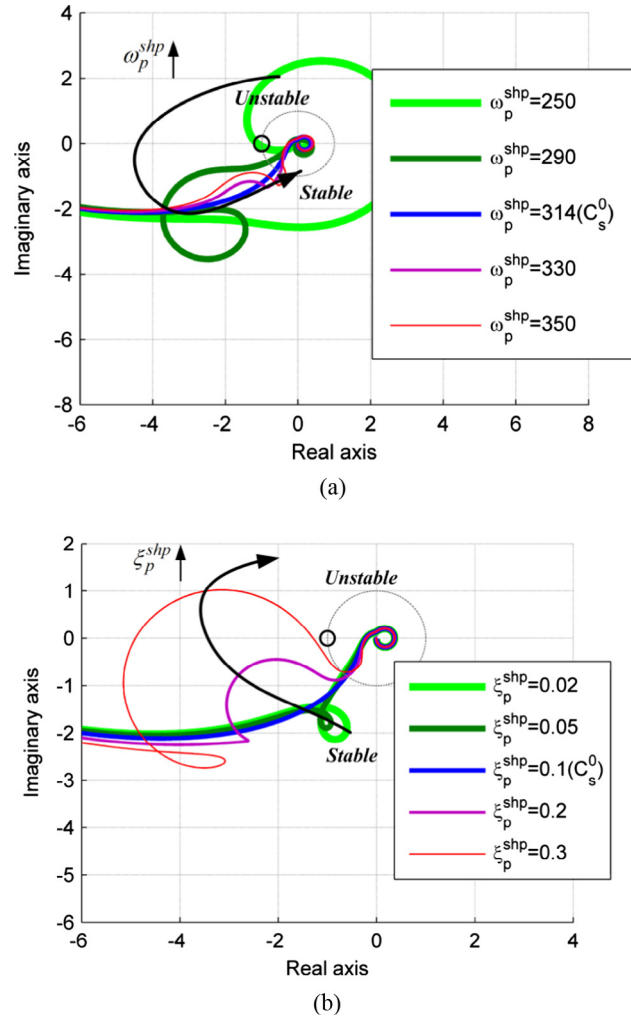


Fig. 5. Nyquist plots and stability situations for shaping filters with different pole parameters. (a) Influences of the pole frequency; (b) Influences of the pole damping ratio.

3.2. Design procedure for robust SDOB

To facilitate the implementation of robust SDOB design for mechatronic motion stages, one feasible design procedure is summarized according to the above discussion. Suppose that the stage model is obtained, and there exists zero/pole resonance within middle frequency range, the following steps are listed:

- Step 1:* Design the feedback controller C that satisfies the required control bandwidth, phase margin, gain margin, etc.
- Step 2:* Design the shaping filter C_s with ideal parameters to cancel the specific resonances in the plant model. Design the nominal model G_n^* that approximates the shaped plant model $\tilde{G}_p$.
- Step 3:* Determine the bandwidth parameter τ for low pass filter Q so that the robust condition for SDOB inner loop in (7) is satisfied. Check the phase margin, gain margin for the overall control loop composed by the feedback controller C and the inner loop model G_{yc} in (2). Try to increase the bandwidth of Q when the robust and stability margin conditions are not violated.
- Step 4:* Adjust the parameter of C_s according to the rule in Section 3.1 to obtain adequate robustness against the uncertainties in the resonances of the plant model. If necessary, Nyquist stability analysis could be performed to help the adjustment of the parameters.
- Step 5:* The feedback controller and SDOB are transformed into control program and run on the control system to test the actual control performances. If the results are not satisfactory, one or more Steps 1–4 should be carried out again to optimize the control performances.

3.3. Summary of the proposed SDOB

Compared with the shaping filter and SDOB in [12], the proposed robust SDOB has the following properties:

1. The optimization method for notch filter in [12] has been abandoned in the proposed SDOB, and zero/pole cancellation method with necessary robust considerations is used for the shaping filter.
2. The parameter is designed according to experiential robust principle without optimization. The utilized experiential robust principle is obtained through the robust analysis of damping ratios, frequencies of zero and pole, which is similar to the experimental design method.
3. The proposed shaping filter is robust to the notch frequencies and damping ratios to certain extent. However, the pole damping ratios and the notch frequencies are directly used in the method in [12], which makes it not robust to the estimation error of damping ratios in practice.

The proposed SDOB method has the following advantages: firstly, the experimental method used in the method make the design of SDOB easily; secondly, four parameters are used in the filter design (notch filter in [12] only uses 3 parameters), which leads to more adjustable freedom; finally, the proposed method is robust to the model parametric variations.

4. SDOB design cases for precision positioning stages

In this Section, the SDOB design procedure is applied to two different mechatronics motion stages to investigate the practical performances of the method.

4.1. Motion control of a linear motor stage

The proposed method is firstly tested on the X-axis of a biaxial linear motor driven stage, which is shown in Fig. 6. The X-axis is composed of an iron core type linear motor (Tecnotion TL Series), a slide guide, a motor driver (Elmo), a linear encoder (resolution 0.15625 μm). The control system is dSpace MicroLabBox DS1202 with Matlab Simulink system.

The first step for loop shaping controller design is to acquire the plant model. Frequency domain identification is performed to obtain the frequency response function (FRF), and a parametric model is then obtained by curve fitting. Fig. 7 shows the Bode plots of the obtained identified FRF and fitted model. There are two zero-pole resonances where one is around 70 Hz and the other is around 600 Hz. Then one manual loop shaping basic feedback controller is designed, and the frequency domain performance specifications are: phase margin 50.2, gain margin 6.1 dB, crossover frequency 114 Hz. As the second resonance at 600 Hz is beyond the control bandwidth, only the first resonance at 70 Hz is considered in SDOB design. The resonance parameters could be obtained from the fitted model: $\omega_z = 71.2$ Hz, $\xi_z = 0.107$, $\omega_p = 78.9$ Hz, $\xi_p = 0.135$. The nominal plant is obtained by discretizing the ideal stage model which comprises mass and damping, i.e., $m\ddot{x} + c\dot{x} = u$ where the parameters are $m = 0.18$ (V/m/s²), $c = 1.1$ (V/m/s) [15–17]. The delay step for the digitalized nominal model is $m = 2$.

In this case, the stage is controlled to track a fourth-order trajectory of which the parameters are: distance 0.2 m, steady velocity 0.2 m/s, maximal acceleration 10 m/s². Fig. 8(a) shows the time plots of position, velocity and acceleration of the trajectory curve. To make the diagram of the trajectory clear, Fig. 8(b) is presented to show the local detail during 0–2 s.

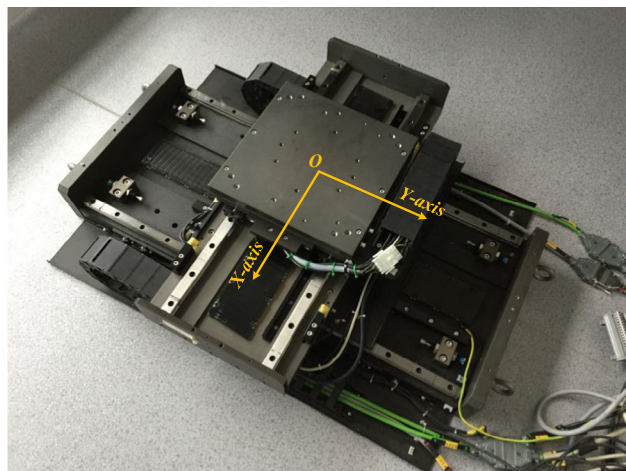


Fig. 6. A linear motor driven stage.

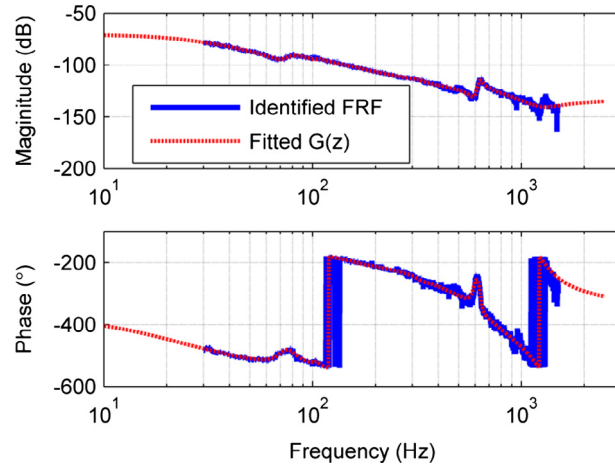


Fig. 7. Bode plots of the identified FRF and fitted model for the linear stage.

Beside the feedback controllers, a constant acceleration and velocity feedforward controller is used to compensate the inertia force and viscous friction during the motion.

Experiments are designed to verify the effectiveness of the robust criteria for the shaping filter parameter in Section 3. Parameter perturbations are applied to a benchmark shaping filter $C_s^0(z)$. As the ideal shaping filter could not stabilize the control loop in actual experiment, some adjustment is made according to the robust design criteria in Section 3. The finally used parameters for $C_s^0(z)$ are $\omega_z^{shp} = 77$ Hz, $\zeta_z^{shp} = 0.2$, $\omega_p^{shp} = 71.2$ Hz, $\zeta_p^{shp} = 0.2$.

Certain perturbation is applied on one of the four parameters of $C_s^0(z)$ in each case, and steady tracking error during the uniform motion segment is calculated. The control cases and results are listed in Tables 1–4. Because damping ratio estimating is usually not very accurate. Cases of rather large variation regions are tested especially for damping ratios. In the experiment results, the influences on stability for each parameter are generally consistent with the developed robust design criteria. One major exception is that instability occurs when $\zeta_z^{shp} = 0.15 > \zeta_p = 0.135$ in Table 2. The possible reason may be the obtained model damping ratio deviates from its actual value (for example actually $\zeta_p = 0.16$ and the rule is violated). Estimating error also may be the reason for instability when $\omega_z^{shp} = \omega_{p,1} = 78.9$ Hz in Table 1. In Table 2 instability occurs when $\zeta_z^{shp} = 0.6$. That is because of the zero damping ratio shifts too much. The similar situation occur for pole damping ratio if Table 4. The experiment result the rule is valid within a small region around the ideal value. But actually the region is enough for the estimating errors, even for damping ratios.

Then one “optimal” shaping filter is redesigned to improve the control performance, and the following parameters are used: $\omega_z^{shp} = 77$ Hz, $\zeta_z^{shp} = 0.25$, $\omega_p^{shp} = 72$ Hz, $\zeta_p^{shp} = 0.2$. Comparative experiments are carried out when the stage is controlled

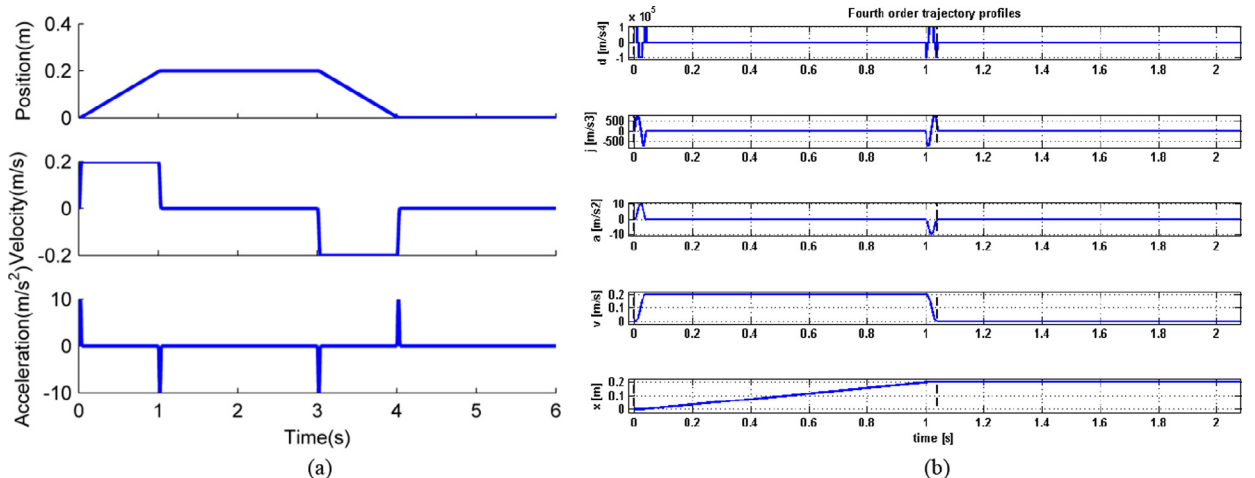


Fig. 8. (a) Position, velocity and acceleration of the trajectory for control experiments. (b) Fourth-order trajectory profiles.

Table 1Experiment results for different shaping zero frequency ω_z^{shp} .

ω_z^{shp} (Hz)	Tracking error (μm)		Stability
	RMS	Peak	
75	0.634	2.344	Stable
77(C_s^0)	0.781	3.906	Stable
78.9	7.637	15.312	Unstable
81	17.174	29.688	Unstable
83	27.341	45.156	Unstable

Table 2Experiment results for different shaping zero damping ζ_z^{shp} .

ζ_z^{shp}	Tracking error (μm)		Stability
	RMS	Peak	
0.1	24.936	42.812	Unstable
0.15	8.309	17.031	Unstable
0.2(C_s^0)	0.743	3.594	Stable
0.25	0.669	2.969	Stable
0.3	0.691	2.500	Stable
0.4	0.899	3.125	Stable
0.6	1.781	5.469	Unstable

Table 3Experiment results for different shaping pole frequency ω_p^{shp} .

ω_p^{shp} (Hz)	Tracking error (μm)		Stability
	RMS	Peak	
67	19.007	32.344	Unstable
69	9.004	18.594	Unstable
71(C_s^0)	0.760	3.281	Stable
73	0.848	3.906	Stable
75	0.655	2.500	Stable

Table 4Experiment results for different shaping pole damping ζ_p^{shp} .

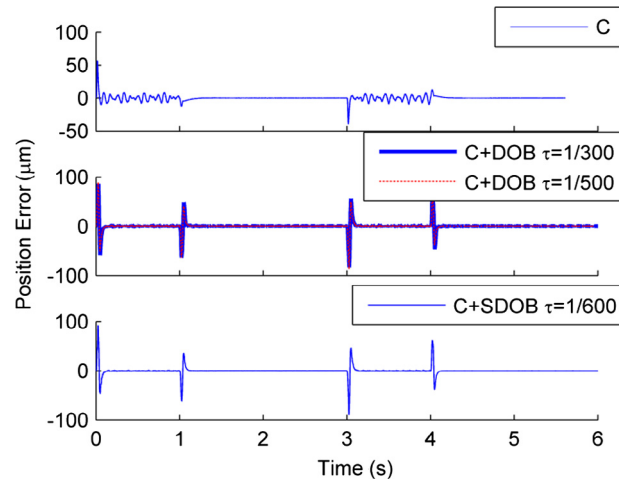
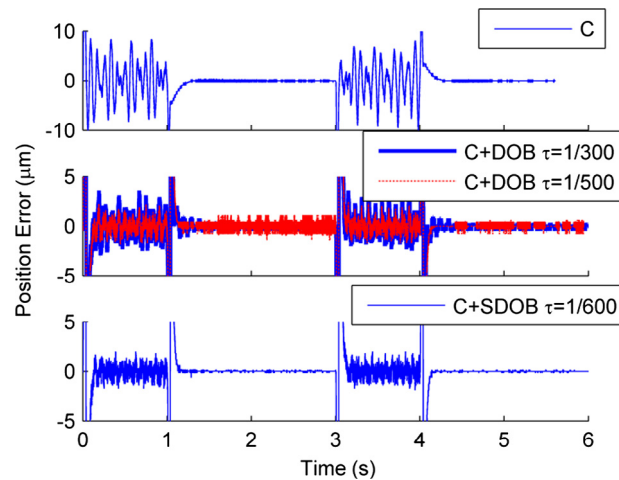
ζ_p^{shp}	Tracking error (μm)		Stability
	RMS	Peak	
0.1	7.198	14.062	Unstable
0.15	0.894	5.937	Stable
0.2(C_s^0)	0.786	3.281	Stable
0.25	1.049	5.156	Stable
0.3	4.345	12.812	Unstable
0.4	14.071	24.219	Unstable

by the basic feedback control (C), basic feedback control plus standard DOB (C + DOB), and basic feedback control plus SDOB (C + SDOB). As there exists large cogging force when the stage moves, the steady tracking error during the uniform motion segments (0.1–1.0s and 3.1–4.0s) is much larger than the positioning error at the stationary positioning segments (1.5–2.8s and 4.5–5.8s). Both the two kinds of errors are calculated to evaluate the performances of different controllers, and results are summarized in Table 5. It seems that DOB reduces the tracking error but increases the positioning error. For basic controller C, the steady tracking errors is 3.894 μm (RMS), and positioning error is 0.132 μm . For C + DOB, when $\tau = 1/300$ is used for Q, the steady tracking error is reduced to 1.157 μm , but the positioning error is increased to 0.214 μm . Further increases the Q bandwidth to the highest ($\tau = 1/500$), the tracking error is reduced to 0.656 μm , but the positioning error is increased to 0.346 μm . For the proposed C + SDOB, $\tau = 1/600$ is used for the Q bandwidth which is larger than that of DOB. The tracking error is 0.628 μm , and the positioning error is 0.036 μm . It is obvious that C + SDOB achieves the best performance. Fig. 9–12 show the detailed time plots and cumulative amplitude spectrum (CAS) plots [14] of the errors with dif-

Table 5

Statistics of the control errors with different controllers.

Controllers		Tracking error (μm)		Positioning error (μm)	
		RMS	Peak	RMS	Peak
C	–	3.894	10.156	0.132	0.469
C + DOB	$\tau = 1/300$	1.157	3.906	0.214	1.094
	$\tau = 1/500$	0.656	2.50	0.346	1.406
C + SDOB	$\tau = 1/600$	0.628	2.50	0.036	0.156

**Fig. 9.** Time plots of tracking error of the stage with different controllers.**Fig. 10.** Enlarged time plots of tracking error with different controllers.

ferent controllers. Herein, for the controller design of this linear motor stage, $\tau = 1/500$ is the maximal bandwidth parameter for Q filter in standard DOB. If the bandwidth is further increased, for example, $\tau = 1/600$, the system will become unstable. In SDOB, a higher bandwidth for Q, i.e., $\tau = 1/600$, could be achieved. C + DOB could enhance the low frequency disturbance rejection ability, but enlarge the effects of noise within the bandwidth of Q filter. Moreover, because of the resonance characteristics, the stability margin could be reduced near the resonance frequency and the system is very sensitive to disturbance or noise around that region. In tracking segment, the cogging force which is the major reason for tracking error mainly locates in low frequency, and therefore could be rejected well by DOB. However, in positioning segment, other disturbance and noise is the major reason for error, and the error becomes large with C + DOB due to the small stability margin around resonance frequency. For C + SDOB, the resonance characteristic is shaped by a filter. Therefore the stability margin around the resonance frequency is enough, and the positioning error is also very small.

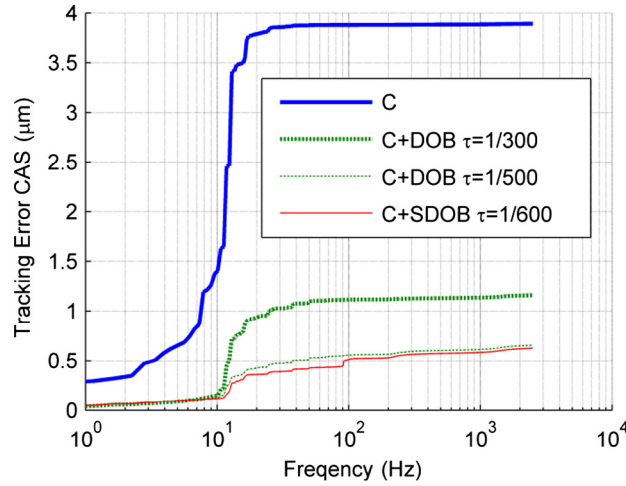


Fig. 11. Cumulative amplitude spectrum (CAS) plots of steady tracking error during uniform motion segments with different controllers.

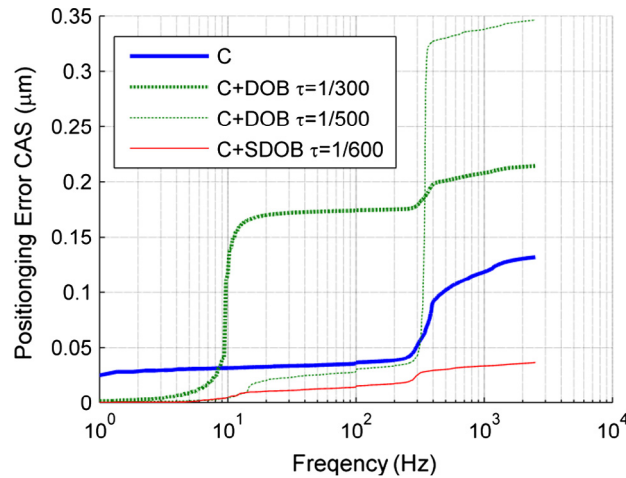


Fig. 12. Cumulative amplitude spectrum (CAS) plots of positioning error during the stationary segments with different controllers.

4.2. Motion control of a planner voice coil motor stage

In this subsection, SDOB controller is designed for a multi-DOF precision positioning stage [18–20]. The stage is driven by planar voice coil motors and supported by air bearing. The travel range is $2 \text{ mm} \times 2 \text{ mm} \times 20 \text{ mrad}$ in $x - y - \theta_z$. The position sensors are laser interferometers with a resolution of 0.62 nm. The control program is run on an industrial VME computer with vxWorks real time operating system, and the sampling frequency is 5 kHz. The control system contains a 16-bit multi-channel D/A card which outputs control voltages to drive motors via linear current amplifiers. The whole system is designed to achieve nanometer scaled positioning accuracy. The overview of the stage is shown in Fig. 13.

The stage dynamics is firstly decoupled into three SISO models corresponding to each DOF through some static transformations. Then controllers are designed for each DOF individually. In this section only motion control of x model is discussed because other models and designed procedures are similar. Fig. 14 shows the Bode plots of the identified frequency response and fitted parametric model $G(z)$. There is a zero/pole resonance around 400 Hz and several resonances above 700 Hz. The fitted model captures the identified model pretty well except the resonances area between 700 and 900 Hz. In the SDOB design, only the resonance at 400 Hz would be canceled by the shaping filter, and from the fitted model $G(z)$ the parameters are $\omega_z = 405 \text{ Hz}$, $\zeta_z = 0.019$, $\omega_p = 411 \text{ Hz}$, $\zeta_p = 0.023$.

A lead-lag feedback controller $C(z)$ is designed according to $G(z)$, and the phase margin is 37.5° , the gain margin is 6.48 dB, and the crossover frequency is 167 Hz. For the low pass filter Q , $\tau = 1/1000$ is used. And the shaping filter $C_s(z)$ is designed as $\omega_p^{shp} = 405 \text{ Hz}$, $\zeta_p^{shp} = 0.05$, $\omega_z^{shp} = 409 \text{ Hz}$, $\zeta_z^{shp} = 0.04$. The damping ratio of the pole of $C_s(z)$ is slightly larger than ζ_z to reduce the output at that frequency. Fig. 15 shows the open loop Nyquist plots for control loop with feedback controller C only, C

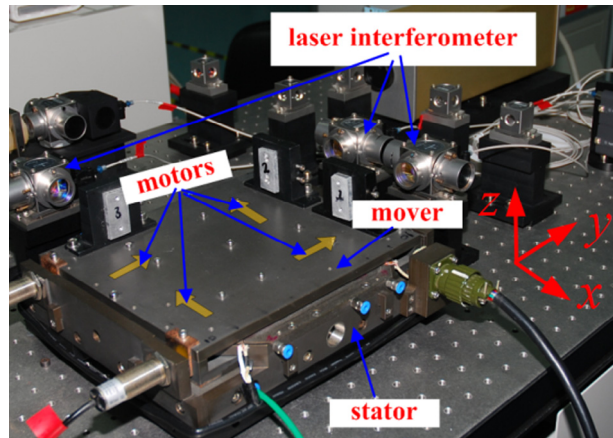


Fig. 13. Overview of a multi-DOF precision positioning stage.

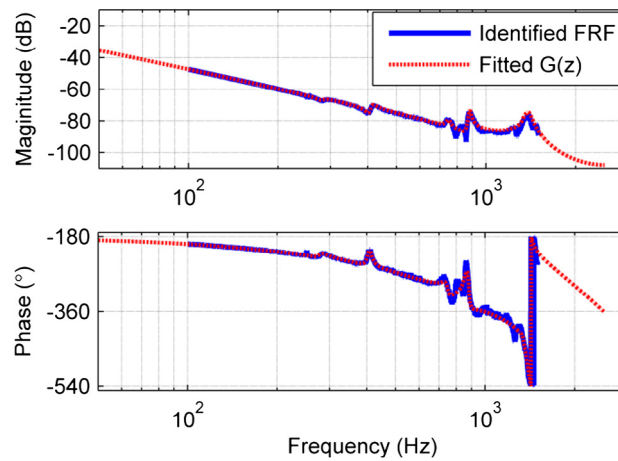


Fig. 14. Identified FRF and fitted model for the x model of the positioning stage.

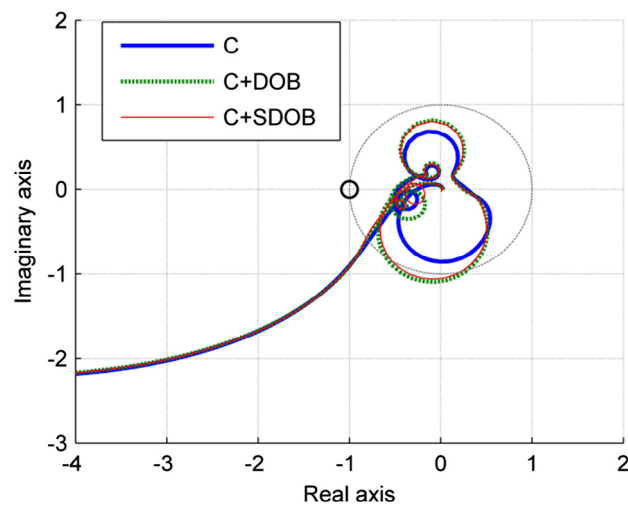


Fig. 15. Open loop Nyquist plots of the control loop for different controllers.

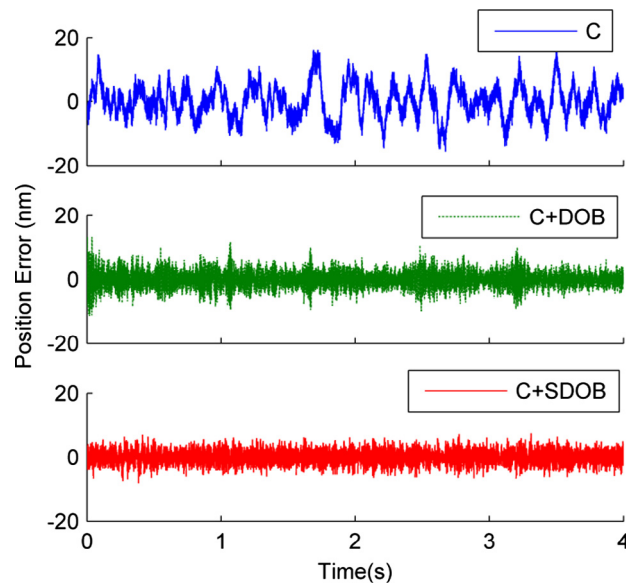


Fig. 16. Time plots of positioning errors for different controllers.

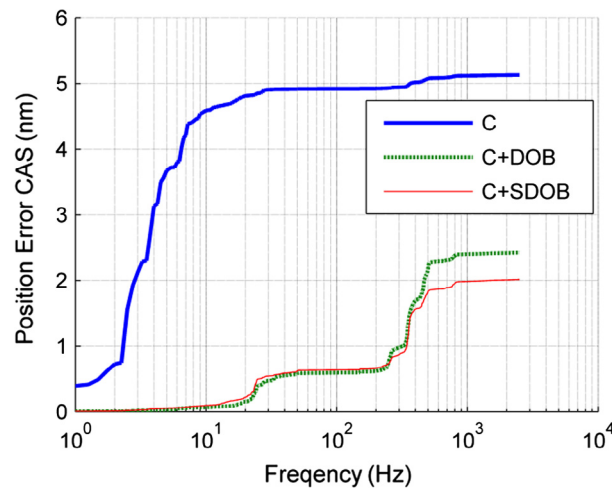


Fig. 17. Cumulative amplitude spectrum (CAS) plots of positioning errors for different controllers.

Table 6

Statistics of the positioning errors with different controllers.

Controllers	Positioning error (nm)	
	RMS	Peak
C	5.1	16.1
C + DOB	2.4	13.3
C + SDOB	2.0	8.1

with DOB and SDOB. DOB and SDOB have the same parameters, and the only difference is that DOB does not have shaping filter. It can be seen that DOB and SDOB almost have no influence on phase margin and gain margin. Fig. 16 shows the time plots of positioning errors, and Fig. 17 shows the cumulative amplitude spectrum plots of the positioning errors. The experiment results show that C + DOB dramatically reduces the positioning error compared with C, and C + SDOB makes further improvements compared with C + DOB. Table 6 shows the statistics of the positioning errors with different controllers. The RMS for the three different controllers: C, C + DOB, C + SDOB, are 5.1 nm, 2.4 nm, 2.0 nm respectively; and the peak val-

ues are 16.1 nm, 13.3 nm and 8.1 nm. It can be seen that, compared to DOB, SDOB suppresses the positioning error around frequency of resonance shaped and improve the positioning accuracy.

5. Conclusions

To facilitate practical engineers to achieve high control performance, a design method of shaping disturbance observer (SDOB) is designed for mechatronic motion plant with middle frequency zero/pole type resonance. Robust criterion for shaping filter design is obtained through frequency domain stability analysis, and the design procedure for SDOB is summarized. Experiments on two different positioning stages show that better control performance can be achieved by the proposed SDOB compared with standard DOB. The proposed method actually provides a simple yet effective way of high precision motion control for motion stages with middle frequency zero/pole type resonance, and also can be applied in industrial applications.

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Appendix A

A virtual plant model is constructed and corresponding controller is designed for the robust analysis of SDOB. Although it is possible to given plant model in discrete form, one continuous model is given first to the make the process easy to understand. The continuous plant model is

$$G_p(s) = e^{-\tau_d s} \frac{1}{s^2} \sum_{i=1}^2 \frac{\frac{s^2}{\omega_{z,i}^2} + 2\zeta_{z,i} \frac{s}{\omega_{z,i}} + 1}{\frac{s^2}{\omega_{p,i}^2} + 2\zeta_{p,i} \frac{s}{\omega_{p,i}} + 1} \quad (\text{A.1})$$

And the parameters are showed in Table A.1. Matched pole-zero method is used to obtained the discrete model without changing the frequencies of resonances, and the resultant model is

$$G(z) = 2.28 \times 10^{-8} z^{-2} \frac{(z+1)(z^2 - 1.984z + 0.9875)}{(z-1)^2(z^2 - 1.978z + 0.9845)} \times \frac{(z^2 - 0.1224z + 0.9415)}{(z^2 - 0.6043z + 0.951)} \quad (\text{A.2})$$

The 1st-order mode is zero/pole resonance and locates at middle frequency; the 2nd-order mode is pole/zero resonance and locates at high frequency beyond control bandwidth.

A feedback controller $C(z)$ and SDOB controller is designed according to the plant model. The feedback controller $C(z)$ has one integrator and one lead, the transfer function is

$$C(z) = 2.43 \times 10^6 \frac{(z - 0.987)(z - 0.974)}{(z - 1)(z - 0.219)} \quad (\text{A.3})$$

The crossover frequency $\omega_c = 1010$ rad/s, phase margin $\text{PM} = 42.3^\circ$, gain margin $\text{GM} = 7.7$ dB. For the design of SDOB, the nominal plant model is

$$G_n(z) = 2 \times 10^{-8} z^{-2} \frac{(z+1)}{(z-1)^2} \quad (\text{A.4})$$

Table A.1

Parameters of the virtual plant for robustness analysis.

Plant parameters		Symbol	Value
1st-order mode	zero	$\omega_{z,1}$	314 rad/s
		$\zeta_{z,1}$	0.1
	pole	$\omega_{p,1}$	390 rad/s
		$\zeta_{p,1}$	0.1
2nd-order mode	zero	$\omega_{z,2}$	7540 rad/s
		$\zeta_{z,2}$	0.02
	pole	$\omega_{p,2}$	6280 rad/s
		$\zeta_{p,2}$	0.02
Delay	time	τ_d	0.0004 s
	step	m	2
Sampling time		T_s	0.0002 s

which is obtained by discretization of plant model without the parts of resonances. The shaping filter $C_s^0(z)$ cancels the 1st-order resonance ideally, i.e., $\omega_z^{shp} = \omega_{p,1}$, $\zeta_z^{shp} = \zeta_{p,1}$, $\omega_p^{shp} = \omega_{z,1}$, $\zeta_p^{shp} = \zeta_{z,1}$. And $\tau = 1/700$ is used for the low pass filter Q in (5) to achieve balanced disturbance rejection and robustness. Then all parts of SDOB could be obtained.

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